

EXPERIMENT - 7

Title

Implement CRUD Operations on the Given Dataset using MongoDB

Aim

To implement **CRUD** operations (**Create, Read, Update, Delete**) on a dataset using **MongoDB**.

Prerequisites

- Basic knowledge of MongoDB
 - Completion of Experiment-6
 - MongoDB Shell (mongosh) installed
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Software Requirements

- Operating System: Windows / Linux / macOS
 - Database: MongoDB Community Server
 - Tool: MongoDB Shell (mongosh)
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Hardware Requirements

- Processor: Intel i3 or above
 - RAM: Minimum 4 GB
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Theory

CRUD stands for:

- **Create** → Insert data
- **Read** → Retrieve data
- **Update** → Modify data
- **Delete** → Remove data

MongoDB stores data as **documents** inside **collections**.

CRUD operations are performed using MongoDB commands on collections.

In this experiment, we use a **Food Delivery dataset** as an example.

Dataset Used

Collection Name: restaurants

Each document contains:

- name
 - location
 - rating
 - cuisine
-

Algorithm

1. Select database.
2. Insert documents (Create).
3. Retrieve documents (Read).
4. Update documents (Update).
5. Delete documents (Delete).
6. Verify operations.

Procedure & Commands

Step 1: Switch to Database

```
use foodDeliveryDB
```

Step 2: Create (Insert Documents)

Insert Single Document

```
db.restaurants.insertOne({  
  name: "Spice Hub",  
  location: "Hyderabad",  
  rating: 4.5,  
  cuisine: "Indian"  
})
```

Insert Multiple Documents

```
db.restaurants.insertMany([  
  {  
    name: "Food Palace",  
    location: "Secunderabad",  
    rating: 4.2,  
    cuisine: "Chinese"  
  },  
  {  
    name: "Green Leaf",  
    location: "Gachibowli",  
    rating: 4.7,  
    cuisine: "South Indian"  
  }  
])
```

Step 3: Read (Retrieve Documents)

View All Documents

```
db.restaurants.find()
```

Pretty Format

```
db.restaurants.find().pretty()
```

Filter Data

```
db.restaurants.find({ location: "Hyderabad" })
```

Step 4: Update (Modify Documents)

Update One Document

```
db.restaurants.updateOne(  
  { name: "Food Palace" },  
  { $set: { rating: 4.6 } }  
)
```

Update Multiple Documents

```
db.restaurants.updateMany(  
  { cuisine: "Indian" },  
  { $set: { delivery: "Available" } }  
)
```

Step 5: Delete (Remove Documents)

Delete One Document

```
db.restaurants.deleteOne({ name: "Spice Hub" })
```

Delete Multiple Documents

```
db.restaurants.deleteMany({ location: "Gachibowli" })
```

Step 6: Verify Data

```
db.restaurants.find().pretty()
```

Sample Output

```
{ acknowledged: true, insertedId: ObjectId("...") }  
{ acknowledged: true, deletedCount: 1 }
```

Expected Result

CRUD operations are successfully performed on the MongoDB collection.

]Result

Thus, CRUD operations on the given dataset were successfully implemented using MongoDB.

Viva Voce Questions

1. What is CRUD?
 2. Difference between insertOne and insertMany?
 3. Difference between updateOne and updateMany?
 4. What does \$set do?
 5. How does MongoDB delete documents?
 6. Can MongoDB handle large datasets?
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Conclusion

This experiment demonstrates how MongoDB efficiently performs CRUD operations on datasets using flexible document-based storage.
